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1  # Full RF Rectifier Report
2
3  ## 1. Optimization Simulation Setup
4  | Item | Value |
5  |-----|-----|
6  | RF Frequency (Hz) | 920 MHz |
7  | RF Input Amplitude (V) | 0.5 |
8  | Temperature (°C) | 27 |
9  | Supply Voltage VDD (V) | 0 |
10 | Process Corner | tt |
11 | Population Size | 50 |
12 | Max Generations | 30 |
13 | Stall Generation Limit | 10 |
14 | Crossover Probability (pcross) | 0.9 |
15 | Mutation Probability (pmut) | 0.2 |
16 | Acceptance Tolerance | 1e-12 |
17 | Stop Cost Threshold | acceptTol |
18 | PVT Corners | tt, ss, ff, sf, fs |
19 | VDD Sweep (V) | 0 |
20 | Temperature Sweep (°C) | -40, 0, 27, 70, 85 |
21 ---
22
23 ## 2. Best Solution Summary
24 | Parameter | Value |
25 |-----|-----|
26 | Generation | 16 |
27 | Index | 19 |
28 | Total Index | 869 |
29 | VOUTMAX | 0.500 V |
30 | POUT | 1.21 µW |
31 | PIN | 1.33 µW |
32 | PCE | 90.69 % |
33 | Sensitivity (PIDBM) | -28.76 dBm |
34 | Ripple (ΔV) | 3.10 mV |
35 | Area | 841 µm2 |
36 | FOM1 | 647.39 |
37 | FOM2 | 18619.4 |
38 | FOM3 | 37207.9 |
39
40 ---
41
42 ## 3. Constraint Verification
43 | Metric | Target | Result | Status |
44 |-----|-----|-----|-----|
45 | VOUT | 0.5-1 V | 0.500 V | PASS |
46 | POUT | ≥1 µW | 1.21 µW | PASS |
47 | PCE | 85-95 % | 90.69 % | PASS |
48 | PIDBM | -35 to -15 dBm | -28.76 | PASS |
49 | Ripple | ≤10 mV | 3.10 mV | PASS |
50 | Area | within bounds | 841 µm2 | PASS |
51
52 ---
53
54 ## 4. PVT Summary (Extremes)
55 - VOUT: **0.476 - 0.519 V**
56 - POUT: **1.10 - 1.30 µW**
57 - PCE: **70.5 - 92.1 %**
58 - PIDBM: **-29.09 to -27.98 dBm**
59 - Ripple: **2.69 - 4.61 mV**
60
61 ---
62
63 ## 5. Monte Carlo Summary
64 | Metric | Mean | Sigma |
65 |-----|-----|-----|
66 | VOUT | 0.500 V | 1.86 mV |
67 | POUT | 1.205 µW | 9.25 nW |
68 | PCE | 90.19 % | 2.10 % |
69 | PIDBM | -28.41 dBm | 0.30 dB |

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70 | Ripple | 3.13 mV | 0.093 mV |
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72 ---
73
74 ## 6. Yield Report
75 - Total Samples: 1000
76 - Passing Samples: 433
77 - **Yield: 43.30%**
78
79 ---
80
81 ## 7. RF Performance Metrics
82 - Dynamic Range (DR): **22.42 dB**
83 - PCE Peak: **90.71 % @ -28.76 dBm**
84 - Sensitivity (@1 V): **-16.05 dBm**
85
86 ---
87
88 ## 8. Impedance Matching (X-Ray)
89 - Input Impedance (pre-match): **1.01 - j148  $\Omega$ **
90 - S11 (before): **-0.036 dB**
91 - S11 (after): **-23.89 dB**
92 - Matched Impedance: ** $\approx 50 \Omega$ **
93
94 ---
95
96 ## 9. Runtime Summary
97 - Total Runtime: **3.43 hours**
98
```